Phuong Huu Vu Tran

Computer Science Undergraduate • Reliable AI • Multimodal Learning

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RESEARCH PROFILE

Computer Science undergraduate at Vietnamese-German University researching reliable AI. My work connects multimodal retrieval, evidence verification, and neuro-symbolic reasoning, with a common focus on making model outputs more trustworthy under noisy evidence, evolving knowledge, and changes in input formulation.

EDUCATION

Vietnamese-German University 2025–2029 (expected)
B.Sc. in Computer Science — Foundation Year GPA: 9.0/10 Ho Chi Minh City, Vietnam

Le Quy Don High School for the Gifted 2022–2025
Specialization in Informatics Khanh Hoa, Vietnam

RESEARCH EXPERIENCE

Undergraduate Research Assistant — PAMI Lab (https://pami-lab.github.io/) Sep 2025–Present
Vietnamese-German University Supervisor: Dr. Cuong Tuan Nguyen

- **FaLCon (co-first author)** [P3]: Co-developed the methodology; designed and ran experiments on data filtering, Qwen3 reranking, and uncertainty-gated consensus; co-wrote and edited the paper. Full system: 95.41% mAP@10 on PAB;
  5th, Track 4 public leaderboard, AI City Challenge, ECCV 2026.
- **GeoRoute** [P4]: Provided supporting ideas and manuscript editing for geometry-aware traffic future-frame prediction.
  5th, Track 5 official leaderboard, AI City Challenge, ECCV 2026.

Undergraduate Research Assistant — RAISE Lab (https://raise-website.vercel.app/) Jan 2026–Present
Ho Chi Minh City University of Technology Supervisor: Dr. Xuan-Bach Le

- **LTM-RAG** [P1]: Co-designed longitudinal trust tracking for poison-resilient RAG; ran most experiments and contributed to manuscript writing and editing.
- **Risk-controlled RAG fact checking (co-first author)** [M1]: Developed the benchmark and conformal Answer/Fallback/Abstain protocol for noisy, updatable evidence; led methodology, experiments, and manuscript writing.
- **CREDENCE (co-first author)** [M2]: Developed Semantic-F1 and a verified claim decomposition and repair framework with convergence analysis; led experiments and manuscript writing.

Undergraduate Research Assistant — Dream AI Lab (https://dream-ai-lab.github.io/) July 2026–Present
Monash University Supervisor: Dr. Trung Le

- **RAVEN (first author)** [M3]: Developed premise-order reasoning and source-to-response certification using executable evidence; led methodology, experiments, and manuscript writing in collaboration with Transformers Lab.

AI Research & Development Intern — Transformers Lab (https://transformerlabs.ai/) Jan 2026–Present
VBI Academy Supervisor: Assoc. Prof. Ngoc-Thanh Dinh

- Contributed to the **xfmr-zem** legal-document RAG pipeline, including corpus construction and end-to-end data processing.

Independent Research July 2026
Paragraph-level argument mining Supervisor: Dr. Hoang Van

- **LLM-INSTRUCT (co-first author)** [P2]: Co-designed metadata-aware retrieval, constrained decoding, and selective debate; ran most experiments and co-wrote and edited the paper.
  1st, UZH Shared Task, ArgMining Workshop, ACL 2026.

PEER-REVIEWED PUBLICATIONS

[P1] Thi-Hong-Cuc Le, **Phuong Huu Vu Tran**, Thuan Mai Duc, and Xuan-Bach Le.
LTM-RAG: Longitudinal Trust Mining for Poison-Resilient Retrieval-Augmented Generation.
IEEE International Conference on Data Mining (ICDM 2026), CORE A*, Research Track
https://lexuanbach.github.io/publication/ICDM2026a.pdf

[P2] **Phuong Huu Vu Tran**[*], Long Minh Vo[*], Son Nguyen Minh Le, and Hoang Van.
LLM-INSTRUCT at UZH Shared Task 2026: Constraint-Aware Retrieval and Selective Debate for Paragraph-Level Argument Mining.
13th Workshop on Argument Mining and Reasoning (ArgMining), ACL 2026
https://aclanthology.org/2026.argmining-1.11/

[P3] Hieu Dinh Trung Pham[*], **Phuong Huu Vu Tran**[*], Thuan Duc Mai, Son Nguyen Minh Le, Khang Le Minh, Hoang Vo, Minh-Chi Phung, Huy Minh Nhat Nguyen, and Cuong Tuan Nguyen.
FaLCon: Facet-Anchored Retrieval with Late Consensus for Sim2Real Text-Based Person Anomaly Search.
AI City Challenge Workshop, ECCV 2026
https://arxiv.org/abs/2608.09474

[P4] Khang Minh Le, Hieu Dinh Trung Pham, Luu Thanh Danh, Nam-Tien Le, Hieu Anh Ngo, **Phuong Huu Vu Tran**, Son Nguyen Minh Le, Nguyen Trong Nghia, Tu Tran Thi Cam, Huy Minh Nhat Nguyen, and Cuong Tuan Nguyen.
GeoRoute: Geometry-Aware Hybrid Inference for Traffic Future-Frame Prediction.
AI City Challenge Workshop, ECCV 2026
https://arxiv.org/abs/2608.09493

[*]Equal contribution.

MANUSCRIPTS UNDER REVIEW

[M1] **Phuong Huu Vu Tran**, Thuan Duc Mai, Hieu Dinh Trung Pham, and Xuan-Bach Le.
Risk-Controlled Evidence-Regime Verification for Noisy, Updatable RAG Fact Checking: A Benchmark and a Conformal Answer/Fallback/Abstain Protocol.
Journal of Artificial Intelligence Research (JAIR) — *Co-first author • Under review*

[M2] **Phuong Huu Vu Tran**, Thuan Duc Mai, and Xuan-Bach Le.
CREDENCE: Claim Reduction for Decomposition & Enhanced Credibility – Semantic Metrics and Convergence Analysis.
Language Resources and Evaluation (LRE) — *Co-first author • Under review*
https://arxiv.org/abs/2606.19819

[M3] **Phuong Huu Vu Tran**, Tinh Pham Phuc Do, Tai Nguyen Khanh, Hoai Minh Phan, Tran Van Nhiem, Linh Ngo Van, Son N. Tran, Ngoc-Thanh Dinh, and Trung Le.
RAVEN: Reasoning Across Premise Orders with Source-to-Response Certification.
International Conference on Learning Representations (ICLR 2027) — *First author • Under review*

SELECTED AWARDS & HONORS

1st Place — UZH Shared Task, ArgMining Workshop
Annual Meeting of the Association for Computational Linguistics (ACL) 2026

5th Place — AI City Challenge
European Conference on Computer Vision (ECCV) 2026

Promising Award (Semifinalist) — Vietnam National Practical AI Competition (AI Thuc Chien)
Vietnam Ministry of Public Security, Vietnam National Data Association & Vietnam Television (VTV) 2026

Bronze Medal — Vietnam Olympiad of Artificial Intelligence (VOAI)
Vietnam Association for Information Processing (VAIP) 2025

First Prize — Khanh Hoa Provincial Science and Engineering Fair
Khanh Hoa Department of Education and Training 2024

TECHNICAL SKILLS

- **Programming & ML:** Python, C++, PyTorch, Hugging Face Transformers
- **Reliable AI & LLMs:** RAG, dense retrieval, reranking, uncertainty calibration, conformal prediction, vLLM
- **Multimodal Learning:** vision-language models, CLIP-based retrieval, video prediction
- **Research & Systems:** Linux, Git, Docker, Slurm, LaTeX/XeLaTeX

LANGUAGES

Vietnamese: Native **English:** IELTS Academic 6.5 (CEFR B2) **German:** Conversational

REFEREES

Trung Le, Senior Lecturer, Department of Data Science & AI (DSAI), Faculty of Information Technology, Monash University, Australia
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Ngoc-Thanh Dinh, Associate Professor of Practice, Center for AI Research (CAIR) & College of Engineering and Computer Science (CECS), VinUniversity, Vietnam
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Google Scholar: https://scholar.google.com.vn/citations?user=Gn4f1boAAAAJ
Website: https://vinuni.edu.vn/vi/people/dinh-ngoc-thanh-2/

Xuan-Bach Le, Lecturer & Head of RAISE Lab, Ho Chi Minh City University of Technology (VNU-HCM), Vietnam
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Google Scholar: https://scholar.google.com/citations?user=FnjHppEAAAAJ
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Cuong Tuan Nguyen, Lecturer and Academic Coordinator, Vietnamese-German University, Vietnam
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Google Scholar: https://scholar.google.com/citations?user=Kx5aOYkAAAAJ
Website: https://pami-lab.github.io/pi/